Kinematic Writing Measures in a Phase II Trial of rTMS for Focal Hand Dystonia

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Overview

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- Why rTMS?

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- Kinematic Writing Assay
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Introduction

What is Dystonia?

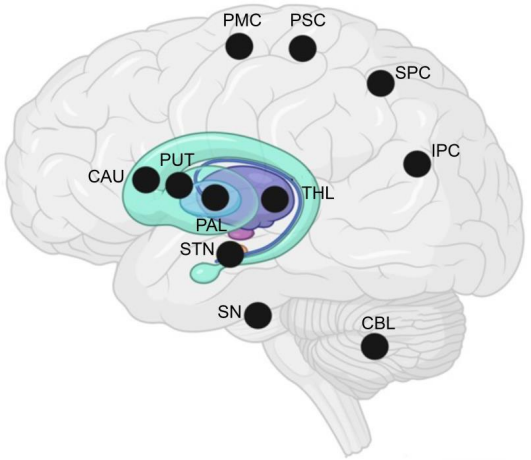


Tanabe *Nature Reviews* 2009

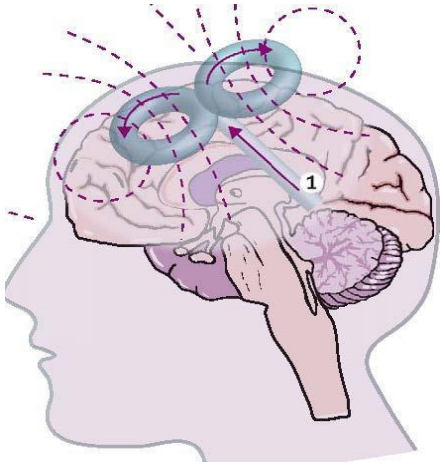
No disease modifying therapy



Motor network



Transcranial magnetic stimulation

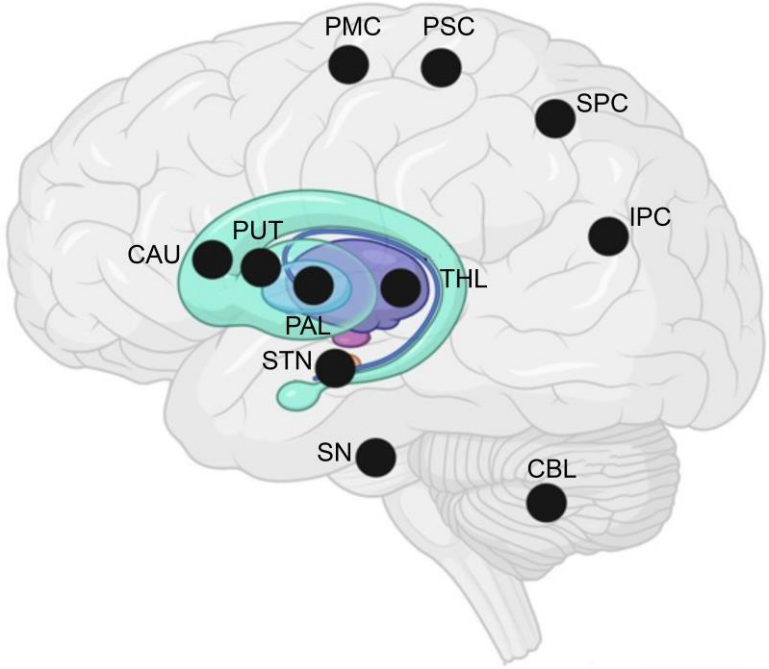


Why rTMS?

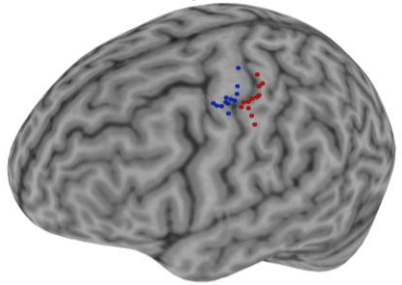
- Pilot study (single session):
  - Improved motor behavior
  - Effect size among highest reported (Bukhari-Parlakturk, Brain Stimulation 2025)

- Current study (accelerated):
  - 4 sessions in one day
  - Tests if more stimulation = better results

- Active TMS = high intensity
- Sham TMS = low intensity

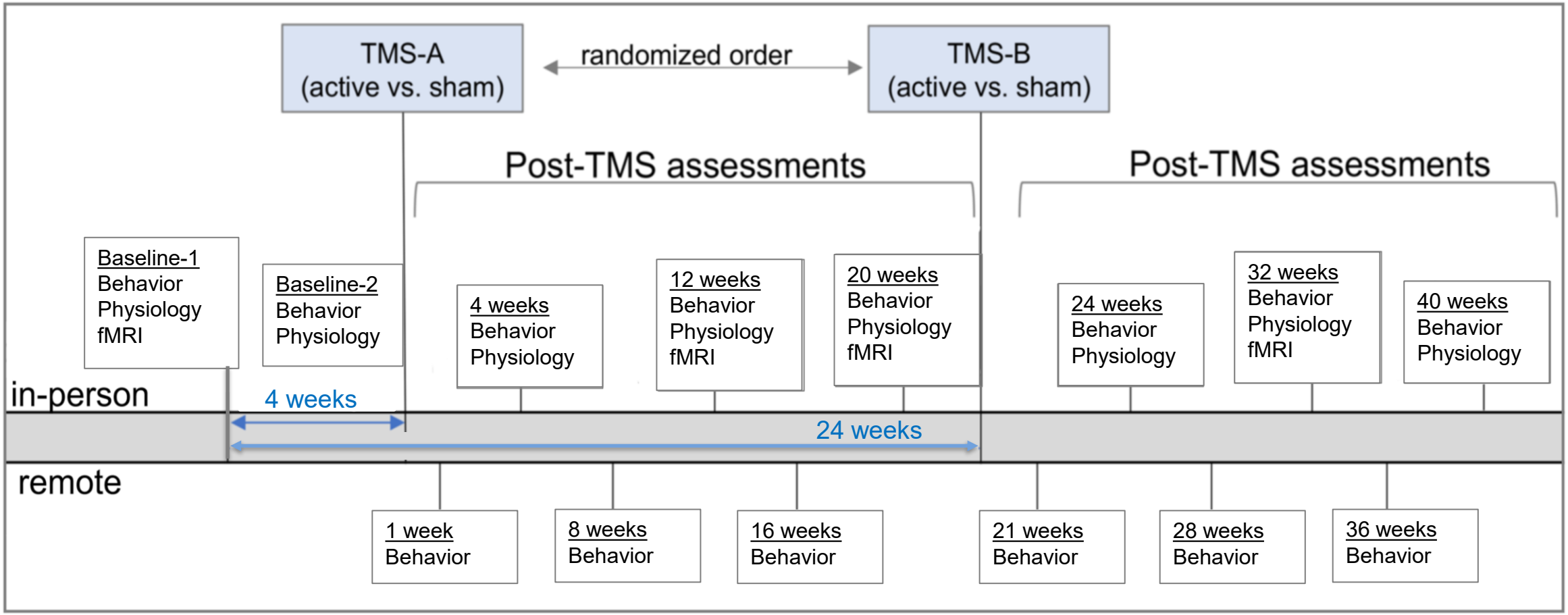


Final PMC and PSC targets

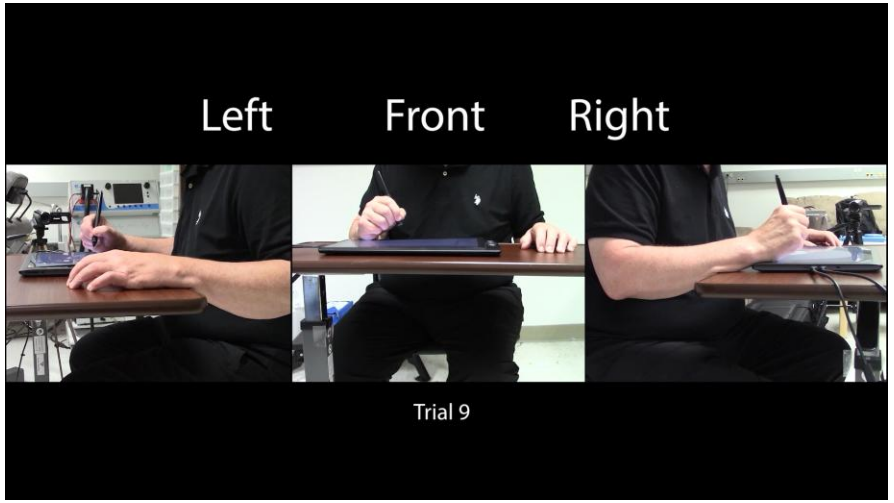


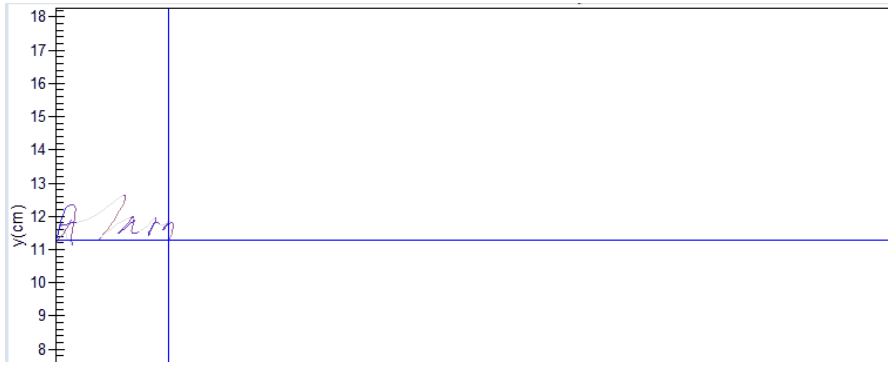
Design & Methodology

Study Design



Kinematic Writing Assay

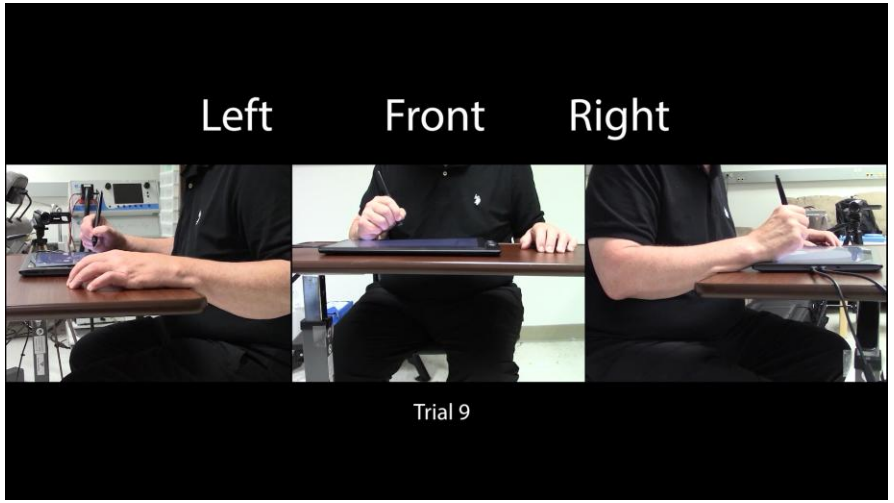


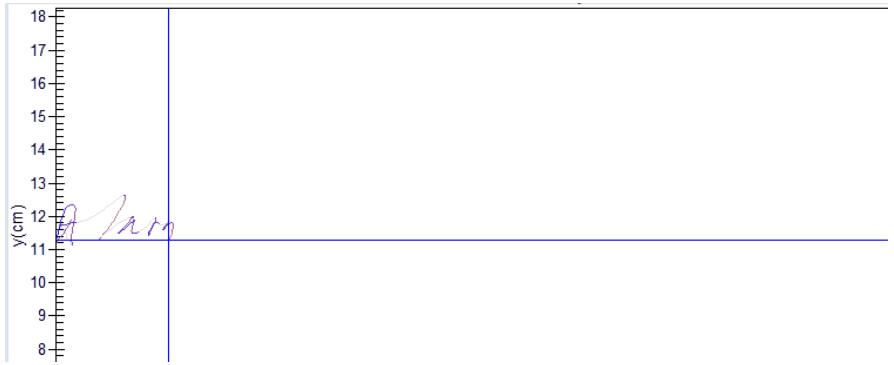


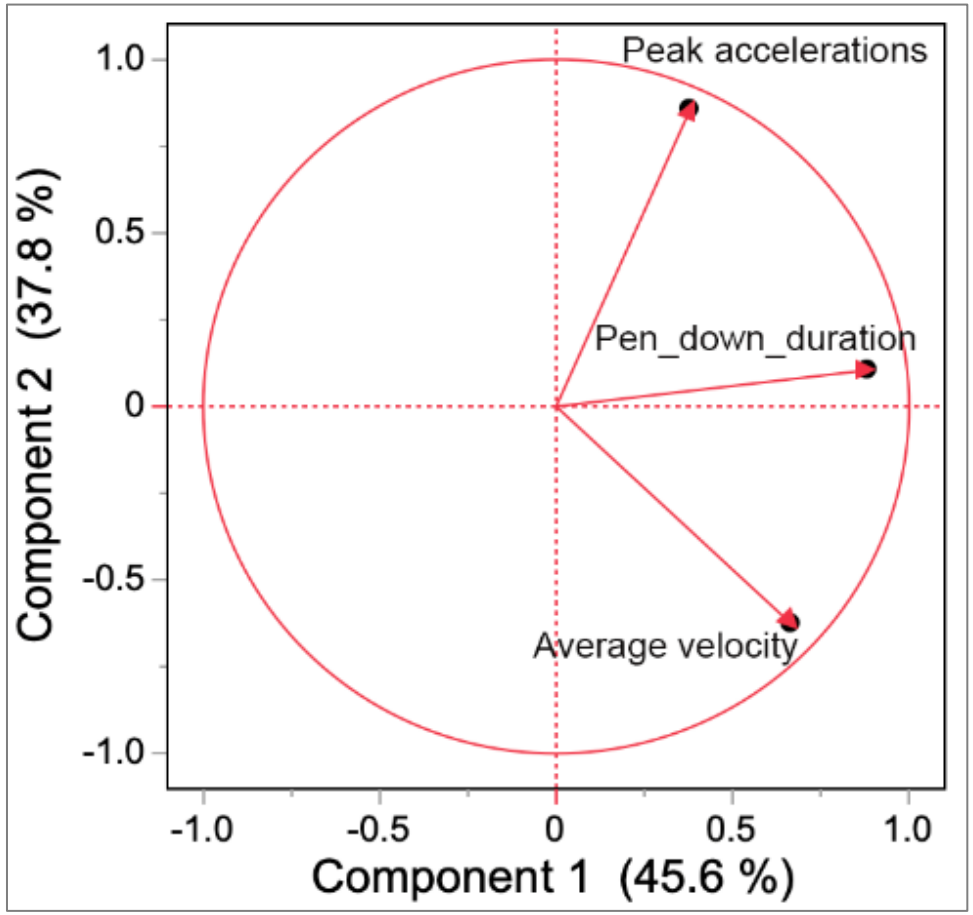
A large fawn jumped quickly over white zinc boxes

Categories	Measures (units)
Dysfluency of Writing	**Peak Accelerations (counts)**
	Score (N)
	Peak Vertical Accelerations (cm/s²)
	Normalized Y-Jerk (a.u.)
	Normalized Jerk (a.u.)
Time of Writing	Duration (s)
	Start Time (s)
	Relative Pen Down Duration (a.u.)
Pressure of Writing	**Average Pen Pressure (N)**
Velocity of Writing	**Peak Vertical Velocity (cm/s)**
	Relative Time to Peak Vertical Velocity (cm/s)
	Average Absolute Velocity (cm/s)
Position of Writing	Start Vertical Position (cm)
	Start Horizontal Position (cm)
Size of Writing	Vertical Size (cm)
	Horizontal Size (cm)
	Absolute Size (cm)
Writing Style	Roadlength (cm)
Angle of Writing	Straightness Error (cm)
	Slant (rad)
	Relative Initial Slant (rad)

Kinematic Writing Assay







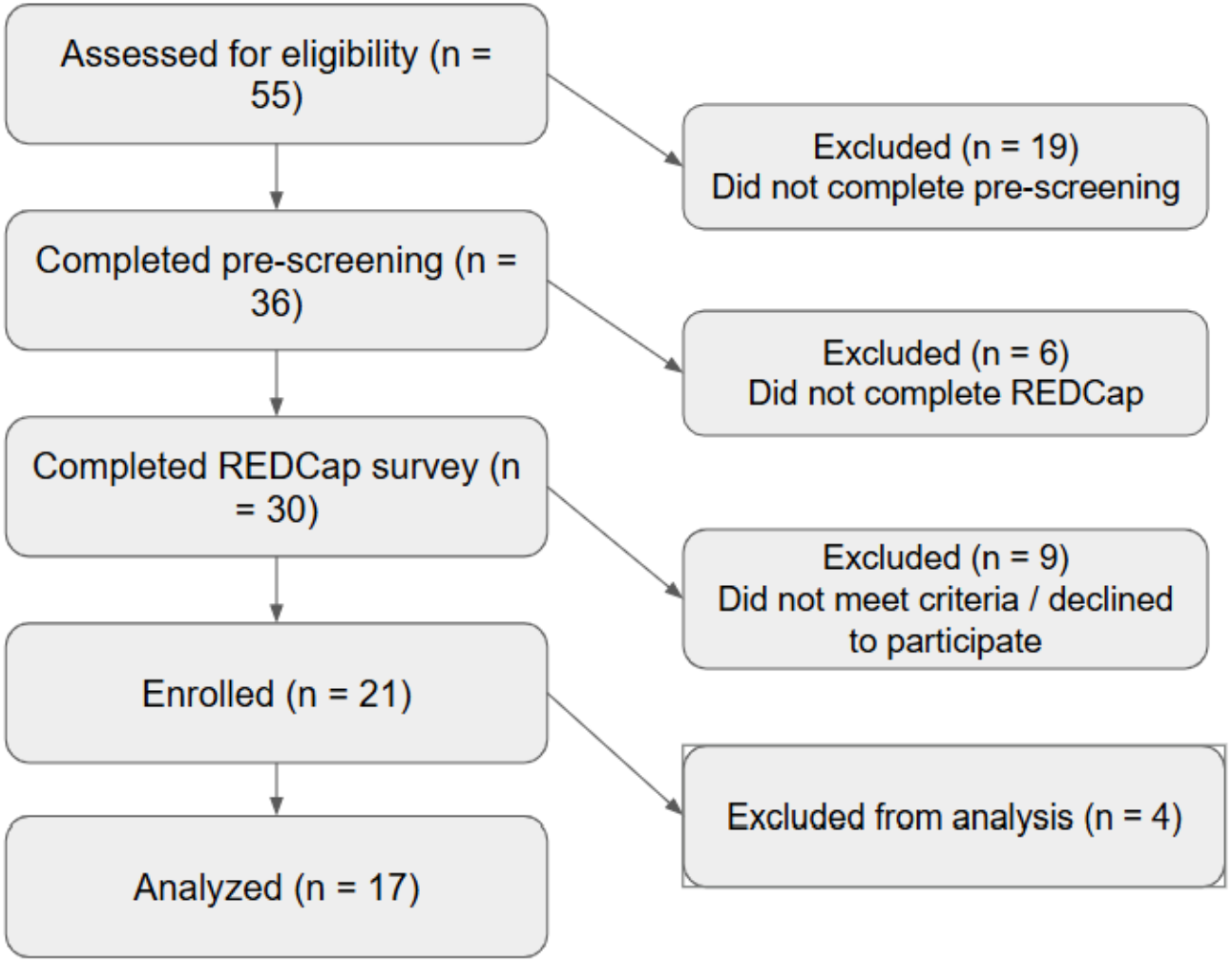
Composite equation = (0.325*peak accelerations) + 0.755*(pen down duration) + (0.569*average velocity) + (0.803*peak accelerations) + (0.098* (pen down duration) – (0.588* average velocity)

Statistical Analysis

- Descriptive & Exploratory Statistics
  - Mean (SD) by arm and timepoint
  - Visualize the data
- Normality
  - Histograms and skewness
- Primary Analysis: Mixed Effects Model
  - Uses all timepoints
- Secondary Analysis: Paired t-test
  - Compares baseline to 20 weeks post-TMS

Results

Study Participants (CONSORT)



Descriptive Statistics

- Arm A improved
- Arm B remained stable

Table 1. Descriptive Statistics for Key Kinematic Measures

Mean (SD) by Timepoint and TMS Arm

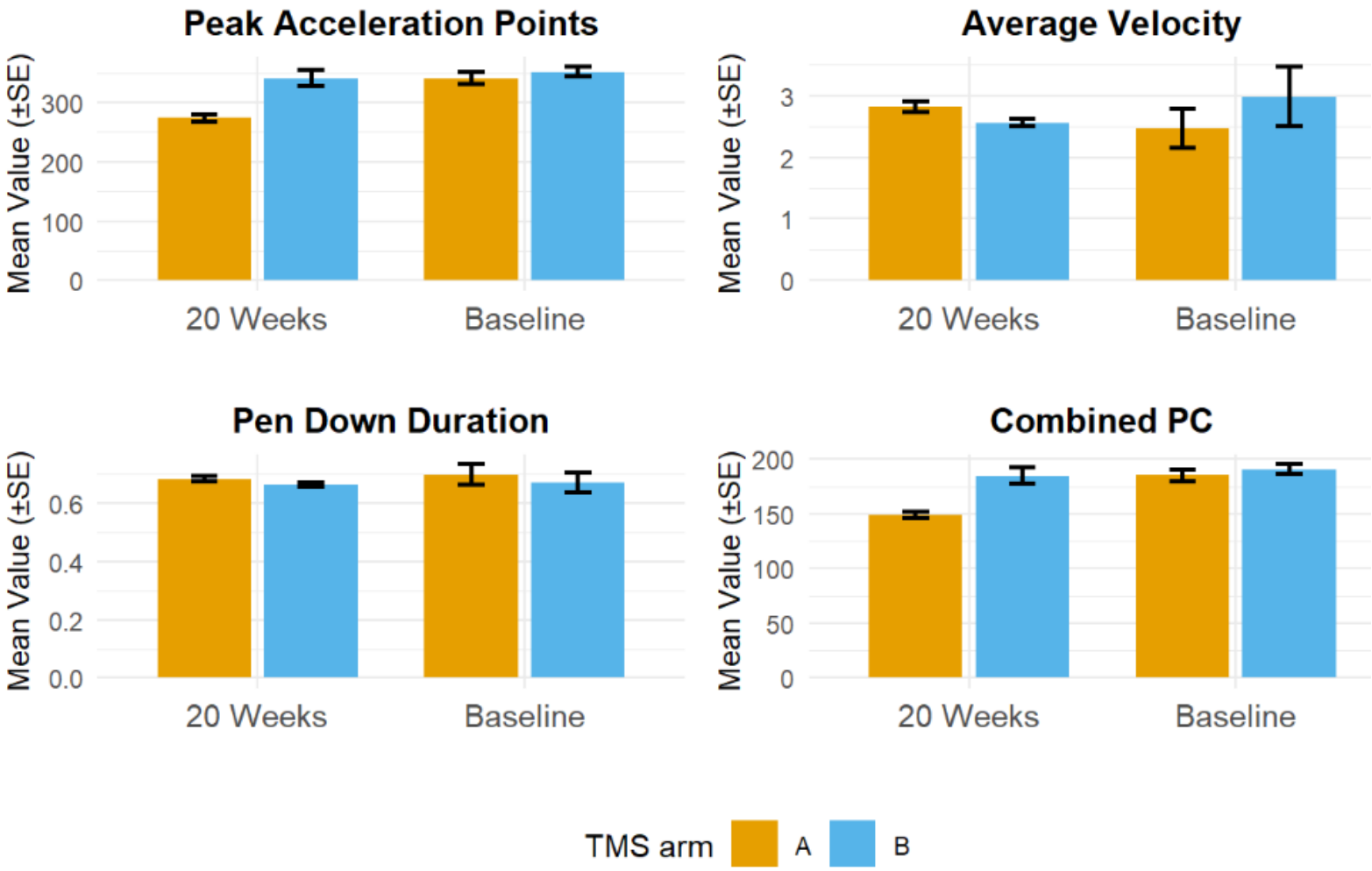
	Arm A	Arm B
Baseline[1]		
Peak Acceleration Points	341.01 (29.39)	351.27 (23.10)
Average Velocity	2.47 (0.88)	2.98 (1.35)
Pen Down Duration	0.70 (0.10)	0.67 (0.10)
Combined PC	185.14 (15.88)	190.70 (12.47)
Post TMS - 4 Weeks		
Peak Acceleration Points	319.33 (110.57)	361.96 (133.50)
Average Velocity	2.86 (1.28)	2.54 (0.49)
Pen Down Duration	0.68 (0.09)	0.66 (0.07)
Combined PC	173.40 (59.85)	196.47 (72.32)
Post TMS - 12 Weeks		
Peak Acceleration Points	308.93 (91.67)	367.63 (139.00)
Average Velocity	2.67 (0.81)	2.57 (0.68)
Pen Down Duration	0.71 (0.07)	0.67 (0.07)
Combined PC	167.78 (49.61)	199.55 (75.30)
Post TMS - 20 Weeks		
Peak Acceleration Points	273.57 (47.30)	340.83 (127.28)
Average Velocity	2.81 (0.75)	2.56 (0.59)
Pen Down Duration	0.68 (0.07)	0.66 (0.06)
Combined PC	148.61 (25.63)	185.03 (68.95)

[1] Baseline represents average of original Baseline and PreTMS timepoints

Exploratory Data Analysis

Figure 1. Baseline vs 20 Weeks Post-TMS

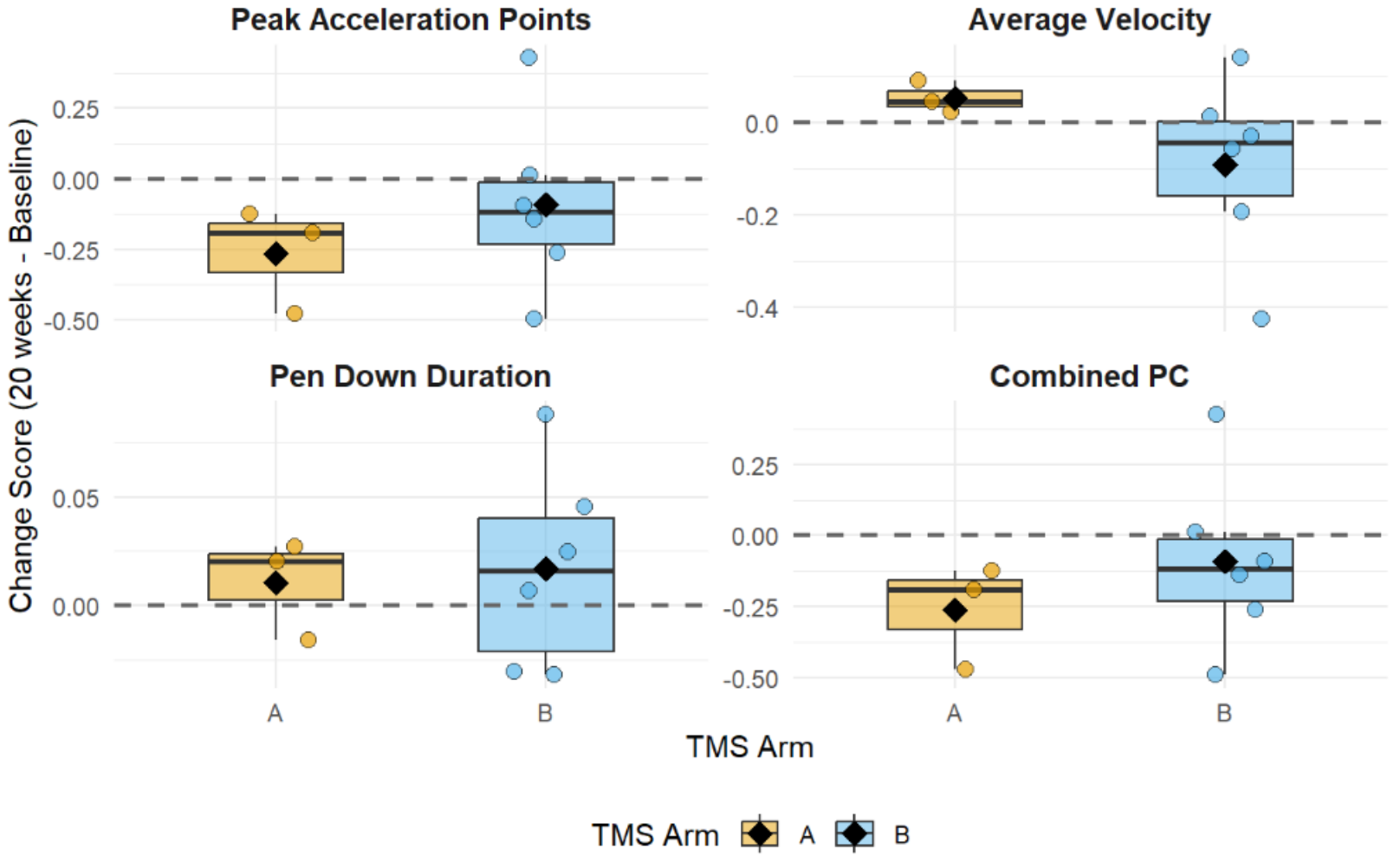
Group mean ± SE



Exploratory Data Analysis

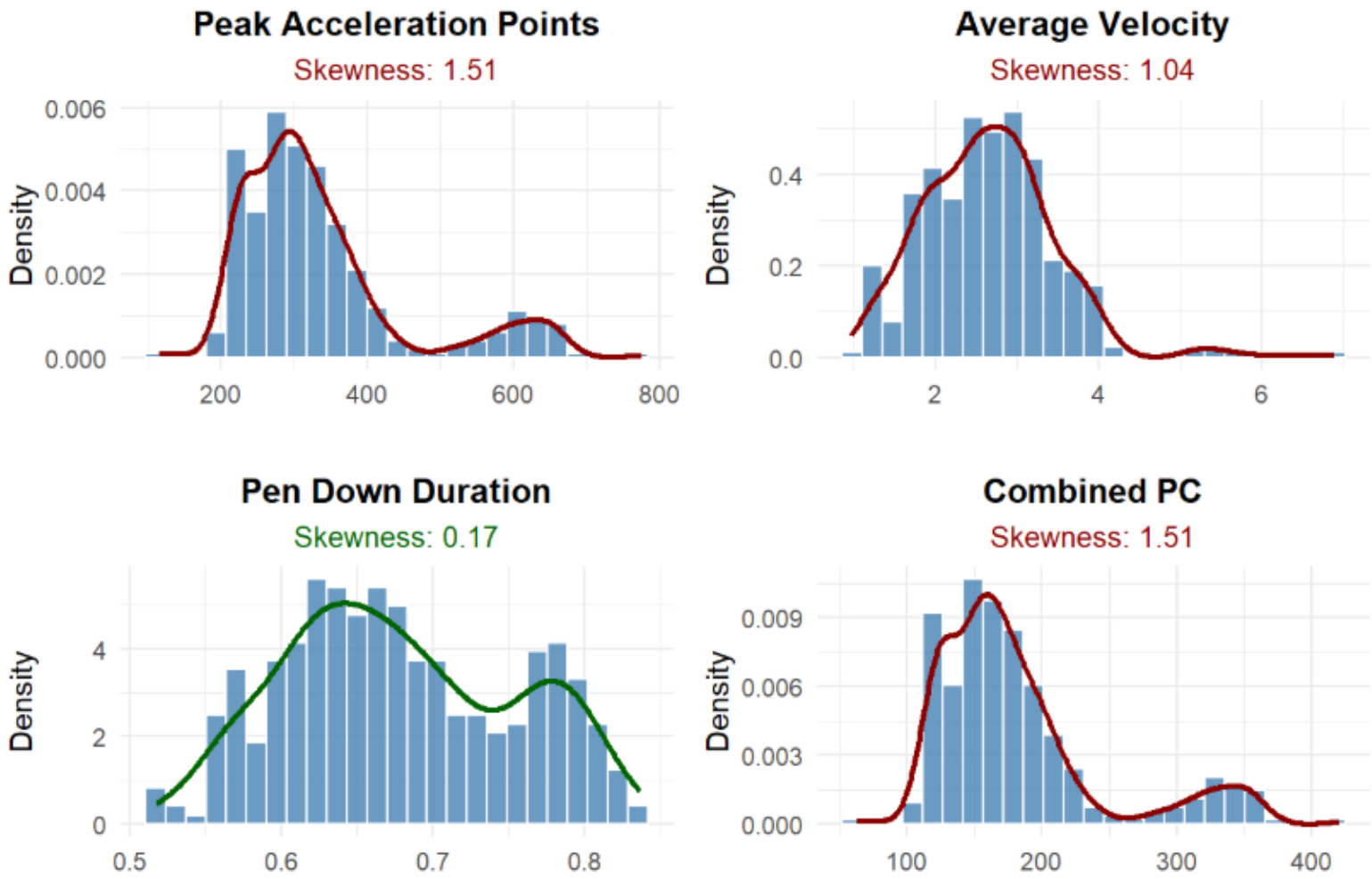
Figure 2. Change Scores from Baseline to 20 Weeks Post-TMS

(+) change = increase; (-) change = decrease | Large diamonds = group mean | Dotted line = no change



Assessing Normality

Figure 3. Distribution of Kinematic Outcome Measures



Mixed Effects Model

Table 2. Mixed Effects Model Results

Effects of TMS on Key Kinematic Measures | Log transformation applied to Peak Acceleration, Average Velocity, and Combined PC[1]

Fixed Effect	Estimate	SE	95% CI	t-value	p-value
Peak Acceleration Points (log)					
Time (weeks)	0.003	0.002	[-0.001, 0.007]	1.557	0.120
TMS Arm (B vs A)	0.139	0.154	[-0.162, 0.441]	0.906	0.383
Time × TMS Arm Interaction	-0.006	0.002	[-0.011, -0.001]	-2.516	<0.05
Average Velocity (log)					
Time (weeks)	0.002	0.001	[-0.000, 0.005]	1.948	0.052
TMS Arm (B vs A)	-0.027	0.130	[-0.281, 0.228]	-0.205	0.841
Time × TMS Arm Interaction	-0.003	0.002	[-0.006, -0.000]	-2.011	<0.05
Pen Down Duration					
Time (weeks)	0.000	0.000	[-0.001, 0.001]	-0.175	0.861
TMS Arm (B vs A)	-0.029	0.043	[-0.113, 0.055]	-0.680	0.509
Time × TMS Arm Interaction	0.001	0.000	[0.000, 0.002]	2.156	<0.05
Combined PC (log)					
Time (weeks)	0.003	0.002	[-0.001, 0.007]	1.556	0.121
TMS Arm (B vs A)	0.139	0.153	[-0.162, 0.439]	0.905	0.384
Time × TMS Arm Interaction	-0.006	0.002	[-0.011, -0.001]	-2.516	<0.05

[1] Models include random intercept and compound symmetry correlation structure

Paired t-test

Table 3. Change Score Differences Between Arms

Independent samples t-test comparing 20-week change scores (Arm A vs Arm B)

Outcome	Mean Difference (A - B)	95% CI	p-value
Peak Acceleration Points (log)[1,2]	[1,2] -0.172	[1,2] -0.572, 0.227	[1,2] 0.336
Average Velocity (log)[1]	[1] 0.144	[1] -0.060, 0.349	[1] 0.133
Pen Down Duration	-0.007	-0.061, 0.048	0.782
Combined PC (log)[1,2]	[1,2] -0.171	[1,2] -0.569, 0.226	[1,2] 0.337

[1] Log transformation applied to Peak Acceleration Points, Average Velocity, and Combined PC

[2] Negative difference indicates greater improvement in Arm A compared to Arm B

Conclusions

Limitations

- Small sample size (n = 17)
- Generalizability
- Multiple outcomes (22 kinematic measures)

Next Steps

- Verify analysis results are correct/reliable
- Perform statistical analysis on the remaining kinematic writing assays
- Include a correction method to decrease Type I error
- Figure how to frame findings for Phase III trial

References

- Bukhari-Parlakturk N, Mulcahey PJ, Lutz MW, Ghazi R, Huang Z, Dannhauer M, Termsarasab P, Scott B, Simsek ZB, Groves S, Lipp M, Fei M, Tran TK, Wood E, Beynel L, Petty C, Voyvodic JT, Appelbaum LG, Al-Khalidi HR, Davis SW, Michael AM, Peterchev AV, Calakos N. Motor network reorganization associated with rTMS-induced writing improvement in writer's cramp dystonia. Brain Stimul. 2025 Mar-Apr;18(2):198-210. doi: 10.1016/j.brs.2025.02.005. Epub 2025 Feb 7. PMID: 39924101; PMCID: PMC11994290.
- Tanabe LM, Kim CE, Alagem N, Dauer WT. Primary dystonia: molecules and mechanisms. Nat Rev Neurol. 2009 Nov;5(11):598-609. doi: 10.1038/nrneurol.2009.160. Epub 2009 Oct 13. PMID: 19826400; PMCID: PMC2856083.